

Recovering Deleted and Damaged Files (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 03

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Time on Task:

2 hours, 39 minutes

Progress:

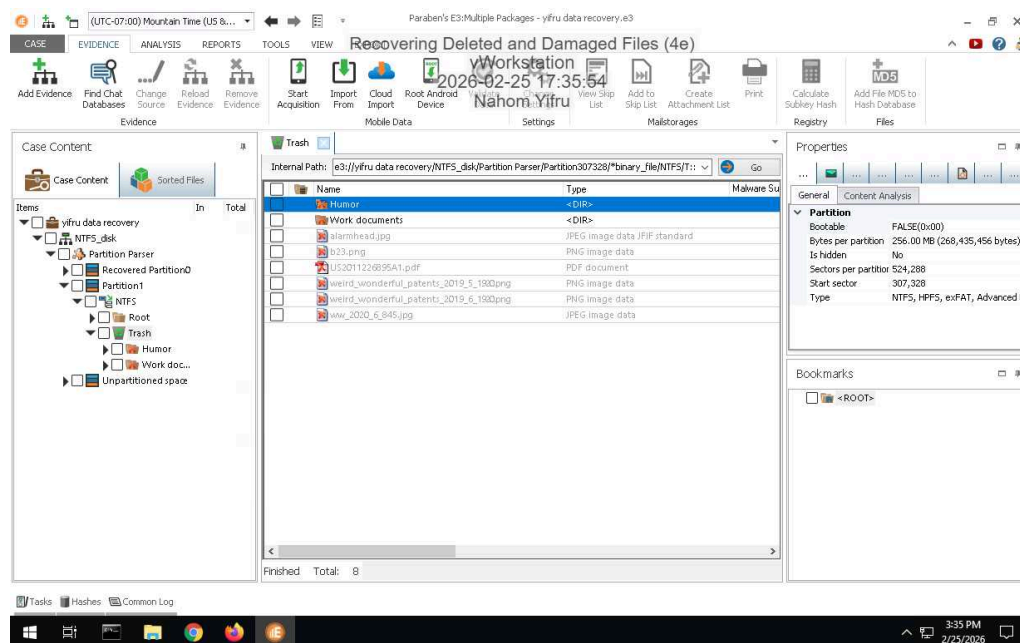
81%

Report Generated: Wednesday, February 25, 2026 at 6:19 PM

Section 1: Hands-On Demonstration

Part 1: Recover Deleted Files from an NTFS Drive Image with E3

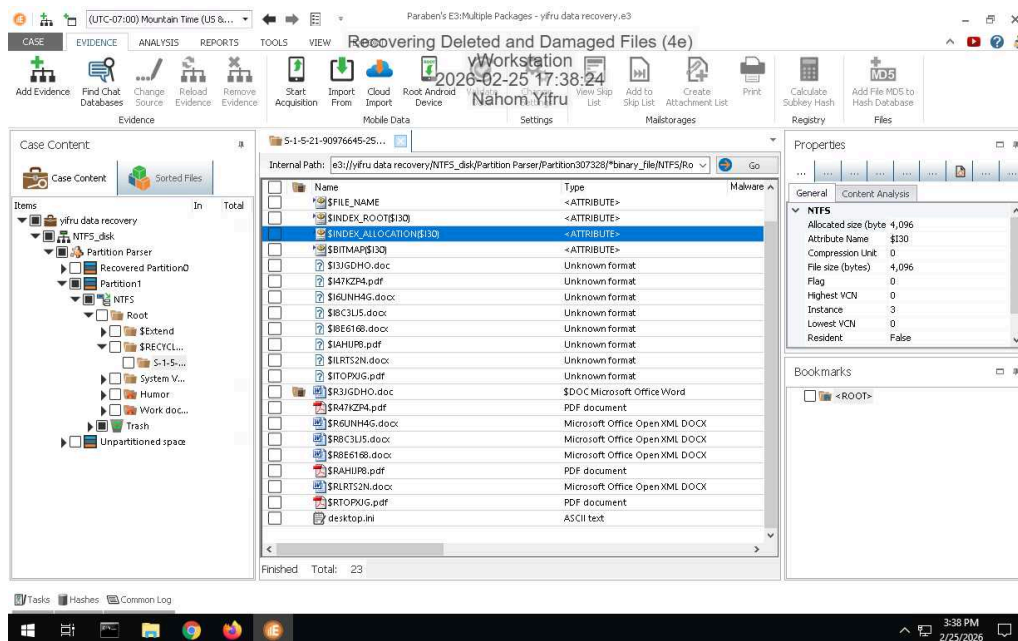
13. Make a screen capture showing the list of recovered files and folders in the E3 Trash folder.



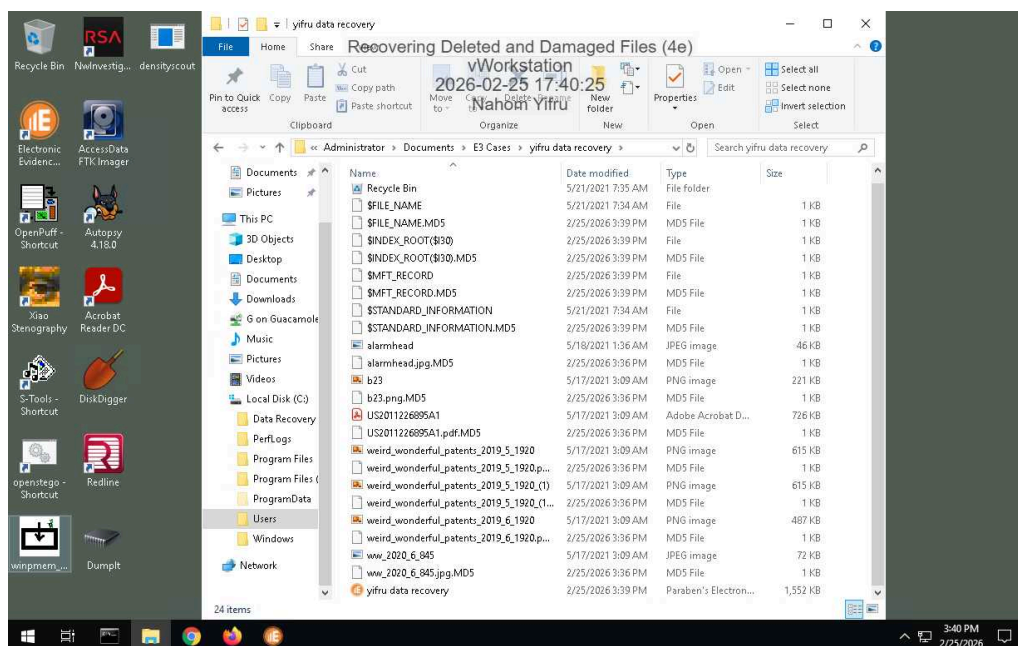
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20. Make a screen capture showing the patent file in the File Viewer.



25. Make a screen capture showing the recovered files in the File Explorer.

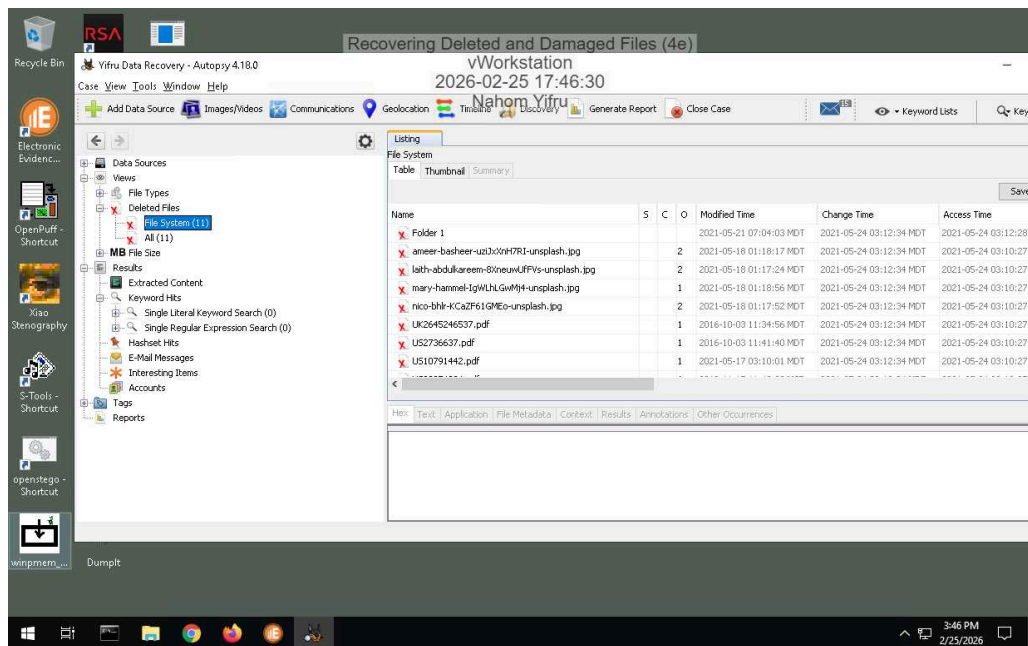


Part 2: Recover Deleted Files from an Ext4 Drive Image with Autopsy

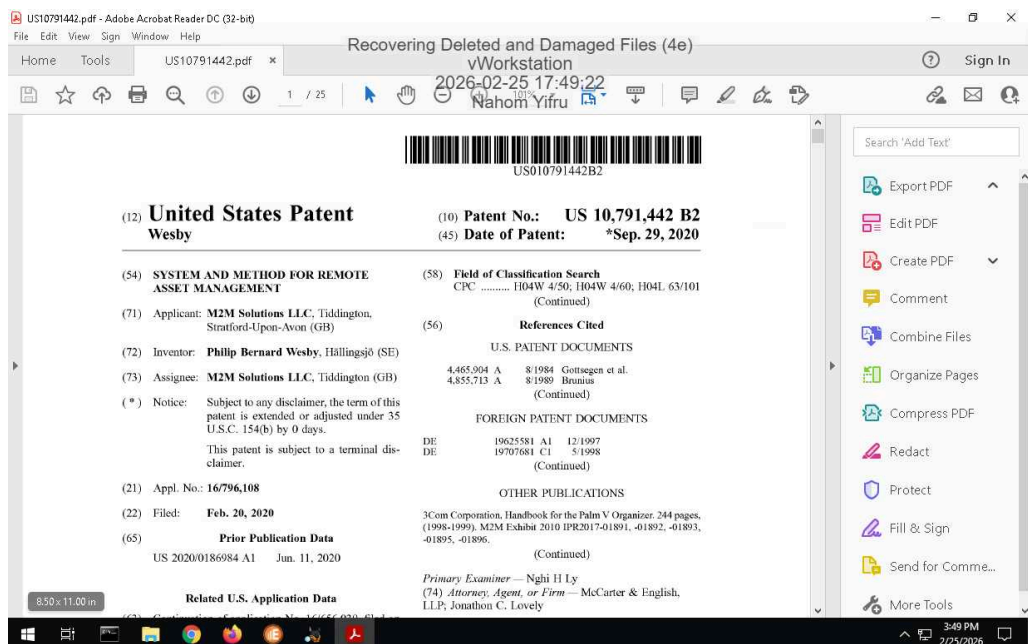
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14. Make a screen capture showing the contents of the list of deleted files in Autopsy.



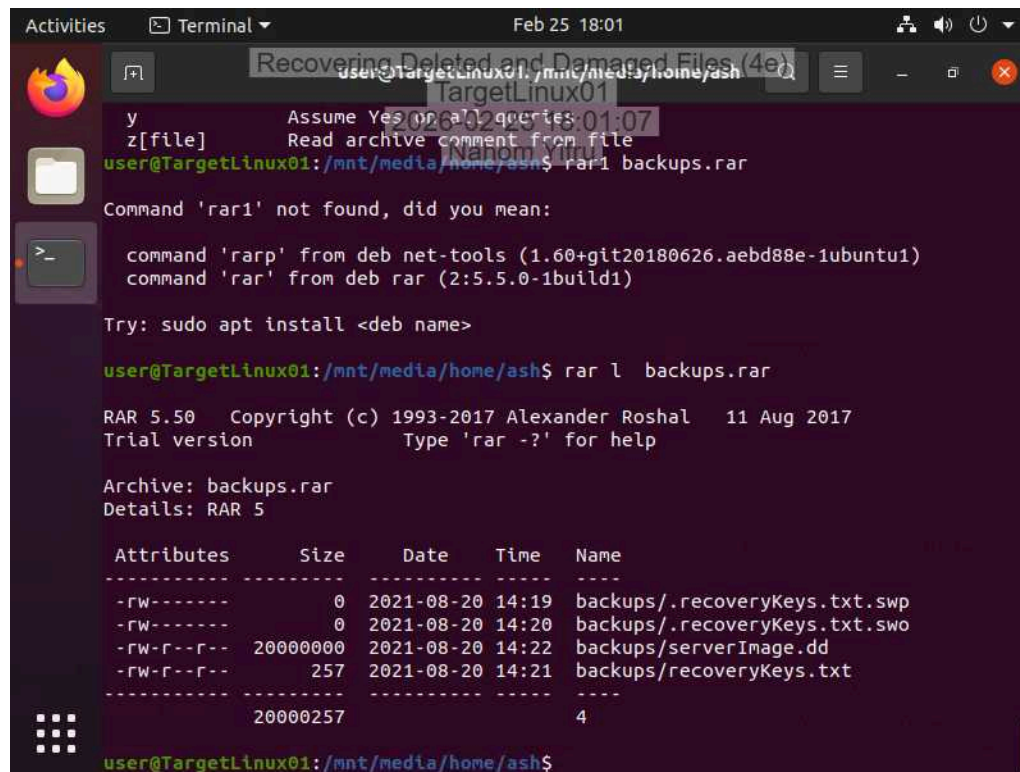
22. Make a screen capture showing the recovered patent file.



Section 2: Applied Learning

Part 1: Recover Deleted Files in Linux with PhotoRec

9. Make a screen capture showing the contents of the RAR archive in the `/mnt/media/home/ash` directory.



The screenshot shows a terminal window on a Linux system. The user is in the directory `/mnt/media/home/ash`. They attempt to run `rarp backups.rar`, which fails with a "Command 'rarp' not found" error. They then try `rarp`, which also fails. The terminal suggests installing the `rar` package using `sudo apt install <deb name>`. After installation, the user runs `rar l backups.rar`, which displays the contents of the RAR archive. The output shows a list of files with their attributes, sizes, dates, times, and names.

```
user@TargetLinux01: /mnt/media/home/ash$ rarp backups.rar
Command 'rarp' not found, did you mean:
  command 'rarp' from deb net-tools (1.60+git20180626.aebd88e-1ubuntu1)
  command 'rar' from deb rar (2:5.5.0-1build1)

Try: sudo apt install <deb name>

user@TargetLinux01: /mnt/media/home/ash$ rar l backups.rar

RAR 5.50 Copyright (c) 1993-2017 Alexander Roshal 11 Aug 2017
Trial version Type 'rar -?' for help

Archive: backups.rar
Details: RAR 5

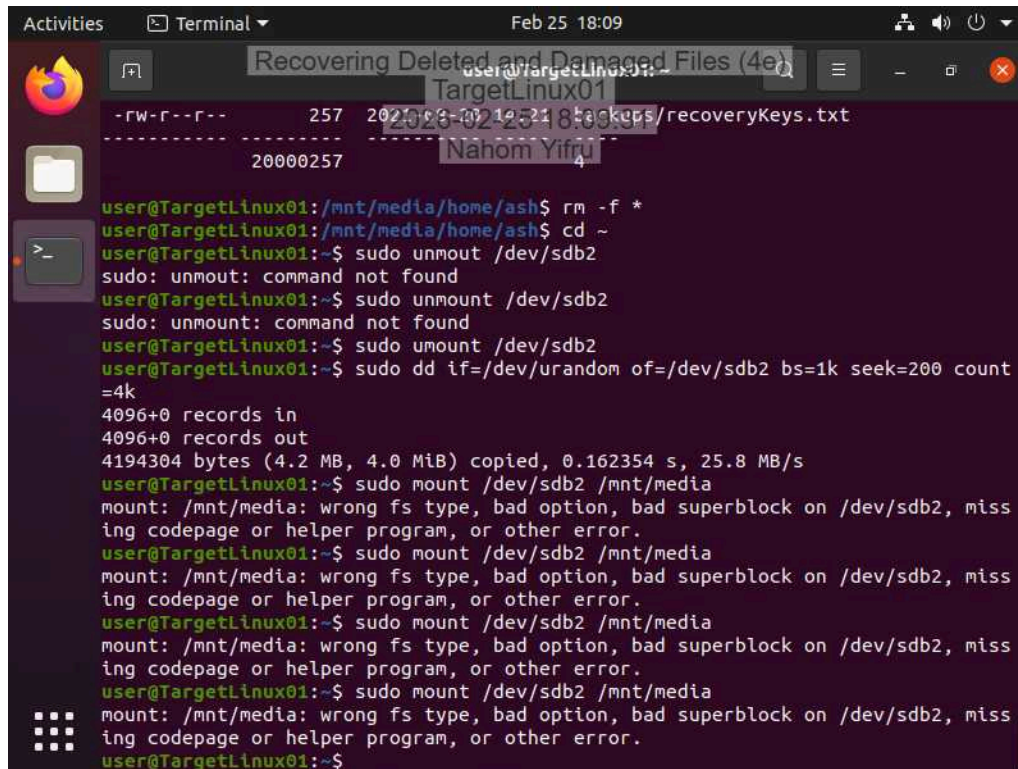
  Attributes      Size      Date      Time      Name
  -----
-rw-----        0  2021-08-20 14:19 backups/.recoveryKeys.txt.swp
-rw-----        0  2021-08-20 14:20 backups/.recoveryKeys.txt.swo
-rw-r--r-- 20000000  2021-08-20 14:22 backups/serverImage.dd
-rw-r--r--    257  2021-08-20 14:21 backups/recoveryKeys.txt
  -----
                20000257                4

user@TargetLinux01: /mnt/media/home/ash$
```


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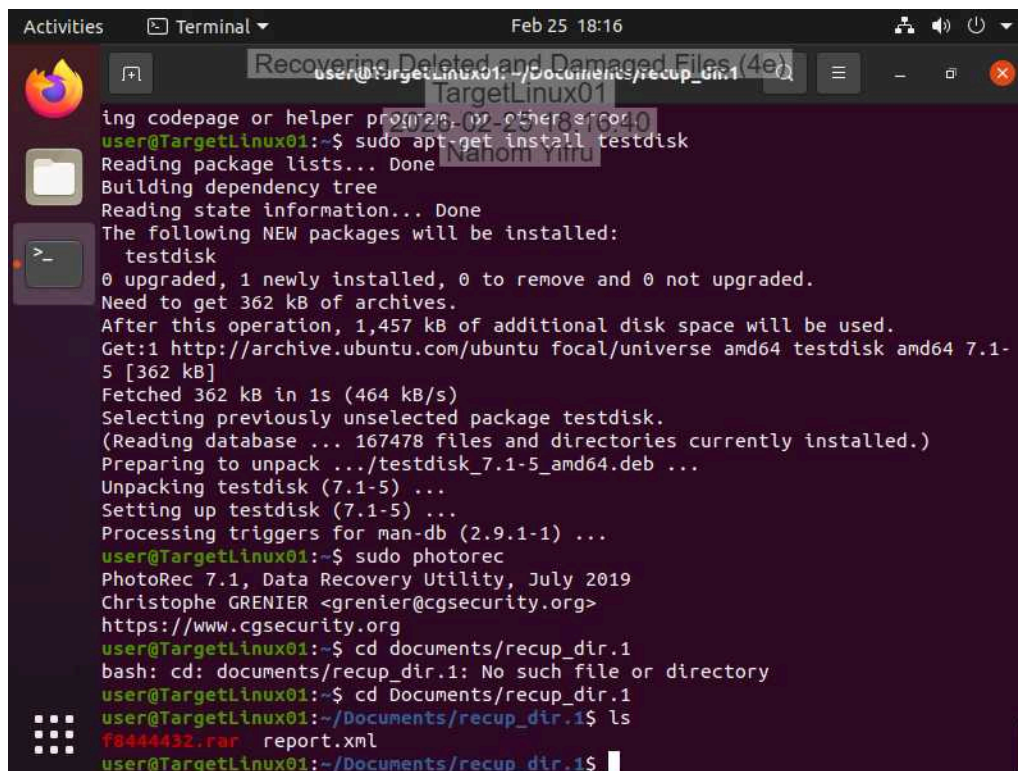
15. Make a screen capture showing the **failed mount attempt on the /dev/sdb2 device.**



A terminal window titled "Recovering Deleted and Damaged Files (4e)" on a system named "TargetLinux01". The terminal shows a series of commands and their outputs. The user attempts to unmount and then mount the device /dev/sdb2. The mount attempts fail with the error: "mount: /mnt/media: wrong fs type, bad option, bad superblock on /dev/sdb2, missing codepage or helper program, or other error." The user also runs a command to create a file on the device using dd.

```
user@TargetLinux01:~/Documents/recoveryKeys.txt
-----
20000257
-----
user@TargetLinux01:/mnt/media/home/ash$ rm -f *
user@TargetLinux01:/mnt/media/home/ash$ cd ~
user@TargetLinux01:~$ sudo umount /dev/sdb2
sudo: umount: command not found
user@TargetLinux01:~$ sudo umount /dev/sdb2
sudo: umount: command not found
user@TargetLinux01:~$ sudo mount /dev/sdb2
user@TargetLinux01:~$ sudo dd if=/dev/urandom of=/dev/sdb2 bs=1k seek=200 count
=4k
4096+0 records in
4096+0 records out
4194304 bytes (4.2 MB, 4.0 MiB) copied, 0.162354 s, 25.8 MB/s
user@TargetLinux01:~$ sudo mount /dev/sdb2 /mnt/media
mount: /mnt/media: wrong fs type, bad option, bad superblock on /dev/sdb2, miss
ing codepage or helper program, or other error.
user@TargetLinux01:~$ sudo mount /dev/sdb2 /mnt/media
mount: /mnt/media: wrong fs type, bad option, bad superblock on /dev/sdb2, miss
ing codepage or helper program, or other error.
user@TargetLinux01:~$ sudo mount /dev/sdb2 /mnt/media
mount: /mnt/media: wrong fs type, bad option, bad superblock on /dev/sdb2, miss
ing codepage or helper program, or other error.
user@TargetLinux01:~$ sudo mount /dev/sdb2 /mnt/media
mount: /mnt/media: wrong fs type, bad option, bad superblock on /dev/sdb2, miss
ing codepage or helper program, or other error.
user@TargetLinux01:~$
```

32. Make a screen capture showing the **compressed files recovered by PhotoRec.**



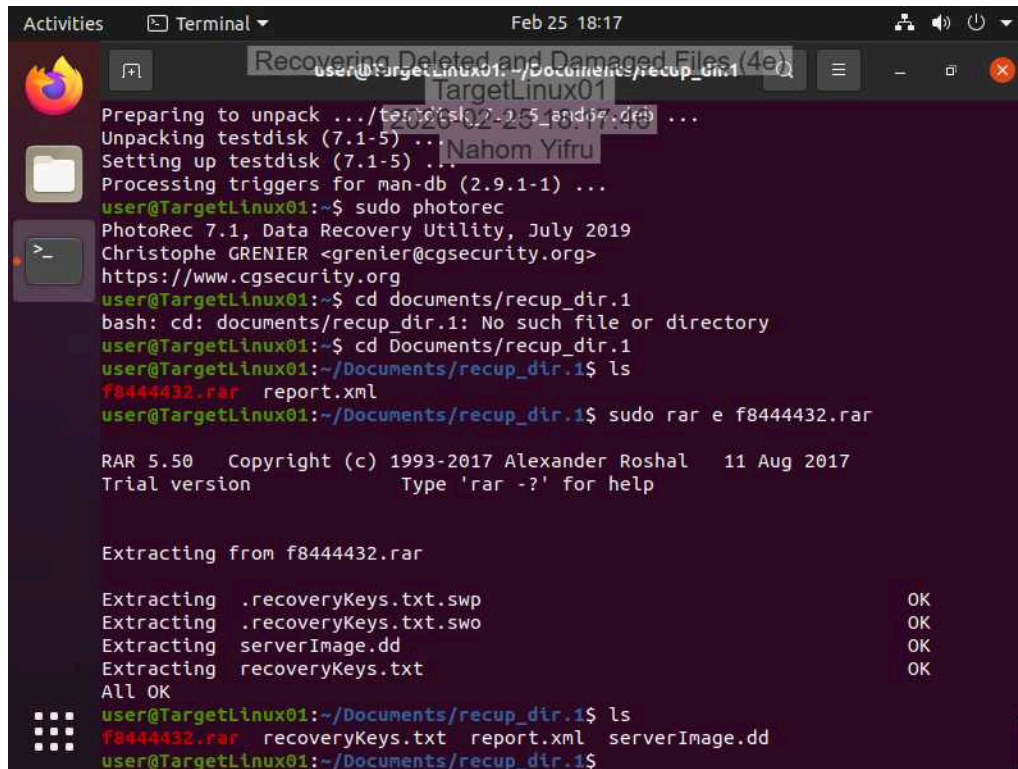
A terminal window titled "Recovering Deleted and Damaged Files (4e)" on a system named "TargetLinux01". The terminal shows the installation of the testdisk package using apt-get. After installation, the user runs the photorec command. The output shows the progress of the recovery process, including the selection of the disk to recover from and the creation of a report file.

```
ing codepage or helper program, or other error.
user@TargetLinux01:~/Documents/recup_dir.1$ sudo apt-get install testdisk
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
 testdisk
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 362 kB of archives.
After this operation, 1,457 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu focal/universe amd64 testdisk amd64 7.1-
5 [362 kB]
Fetched 362 kB in 1s (464 kB/s)
Selecting previously unselected package testdisk.
(Reading database ... 167478 files and directories currently installed.)
Preparing to unpack .../testdisk_7.1-5_amd64.deb ...
Unpacking testdisk (7.1-5) ...
Setting up testdisk (7.1-5) ...
Processing triggers for man-db (2.9.1-1) ...
user@TargetLinux01:~$ sudo photorec
PhotoRec 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
user@TargetLinux01:~$ cd documents/recup_dir.1
bash: cd: documents/recup_dir.1: No such file or directory
user@TargetLinux01:~$ cd Documents/recup_dir.1
user@TargetLinux01:~/Documents/recup_dir.1$ ls
f8444432.rar  report.xml
user@TargetLinux01:~/Documents/recup_dir.1$
```

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35. Make a screen capture showing the backup files recovered from the RAR archive.



The screenshot shows a terminal window on a Linux system. The user is at the prompt `user@TargetLinux01:~/Documents/recup_dir.1`. The terminal output shows the following commands and results:

```
Preparing to unpack .../testdisk-7.1-5-amd64.deb ...
Unpacking testdisk (7.1-5) ...
Setting up testdisk (7.1-5) ...
Processing triggers for man-db (2.9.1-1) ...
user@TargetLinux01:~$ sudo photorec
PhotoRec 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
user@TargetLinux01:~$ cd documents/recup_dir.1
bash: cd: documents/recup_dir.1: No such file or directory
user@TargetLinux01:~$ cd Documents/recup_dir.1
user@TargetLinux01:~/Documents/recup_dir.1$ ls
f8444432.rar  report.xml
user@TargetLinux01:~/Documents/recup_dir.1$ sudo rar e f8444432.rar

RAR 5.50 Copyright (c) 1993-2017 Alexander Roshal 11 Aug 2017
Trial version Type 'rar -?' for help

Extracting from f8444432.rar

Extracting .recoveryKeys.txt.swp OK
Extracting .recoveryKeys.txt.swo OK
Extracting serverImage.dd OK
Extracting recoveryKeys.txt OK
All OK
user@TargetLinux01:~/Documents/recup_dir.1$ ls
f8444432.rar  recoveryKeys.txt  report.xml  serverImage.dd
user@TargetLinux01:~/Documents/recup_dir.1$
```

Section 3: Challenge and Analysis

Part 1: Recover Deleted Files from a FAT Drive Image

Make a screen capture showing the **patent file recovered from the FAT32 drive image within E3.**

Incomplete

Part 2: Recover Deleted Files from a APFS Drive Image

Make a screen capture showing the **patent file recovered from the APFS drive image within Autopsy.**

Incomplete